Requirements Gathering

3.1 Introduction

Requirements gathering is the process of identifying and documenting the business needs, expectations, and technical requirements of a project before implementation.

For the HR Analytics & Employee Attrition Analysis project, this phase focuses on understanding the information HR management needs to monitor workforce performance, investigate employee attrition, and identify potential retention challenges.

The gathered requirements will guide the data preparation process, analytical approach, and development of the interactive Power BI dashboard.

The main goal is to ensure that the final solution addresses the defined business problems and provides meaningful insights to support HR decision-making.

3.2 Stakeholder Identification

The primary stakeholders are the individuals or groups who will benefit from or interact with the project outputs.

Stakeholder	Role & Expectations
HR Management	Monitor workforce size, employee attrition, and retention challenges.
HR Analysts	Analyze employee data, identify patterns, and generate insights.
Department Managers	Understand employee distribution, satisfaction, and performance within their departments.
Senior Management	Use summarized HR insights to support strategic workforce decisions.
Project Team	Collect, prepare, analyze, and visualize the available HR data.

3.3 Functional Requirements

Functional requirements describe the specific features and capabilities that the Power BI solution must provide.

The dashboard should support the following functions:

FR-01: Workforce Monitoring

- Display total employee headcount.
- Show employee distribution by department and job role.
- Analyze workforce demographics, including age and gender.

- Display average employee age and other relevant workforce metrics.

FR-02: Attrition Analysis

- Calculate total employees who left the company.
- Calculate the overall attrition rate.
- Compare attrition rates across departments and job roles.
- Analyze attrition by age group, gender, and employee tenure.

FR-03: Satisfaction Analysis

- Display job satisfaction, environment satisfaction, and relationship satisfaction.
- Compare satisfaction levels between employees who stayed and those who left.
- Analyze work-life balance across different employee groups.

FR-04: Compensation & Working Conditions

- Analyze salary distribution across job roles and departments.
- Compare attrition rates across salary groups.
- Examine overtime and business travel frequency.
- Analyze distance from home and other available working-condition indicators.

FR-05: Performance & Career Development

- Analyze performance ratings across departments and job roles.
- Compare employee self-ratings with manager ratings.
- Examine training frequency and performance.
- Analyze promotions and salary increases.
- Identify employees with missing performance evaluation records.

FR-06: Interactive Analysis

- Provide slicers and filters for relevant employee attributes.
- Allow users to explore different workforce segments.
- Enable comparisons across departments, job roles, age groups, and attrition status.
- Ensure that visualizations respond correctly to user selections.

3.4 Non-Functional Requirements

Non-functional requirements describe the quality standards and operational conditions expected from the dashboard.

NFR-01: Usability

- The dashboard should have a simple and user-friendly interface.
- Visualizations should be easy to understand.

- Information should be organized into clearly defined sections.

NFR-02: Accuracy

- KPI calculations must be accurate and consistent.
- Data relationships must be validated.
- Attrition rates must be calculated using the correct employee population.

NFR-03: Performance

- Dashboard pages should load and respond efficiently.
- Visual interactions and filters should perform smoothly.
- The data model should be organized to avoid unnecessary complexity.

NFR-04: Consistency

- All dashboard pages should follow a consistent design.
- Colors, fonts, chart styles, and KPI formats should be standardized.
- Metrics should maintain consistent definitions across the report.

NFR-05: Maintainability

- Measures and data transformations should be organized and documented.
- The dashboard should allow future updates when new employee data becomes available.

NFR-06: Data Privacy

- Employee information should be handled responsibly.
- Sensitive employee details should not be unnecessarily exposed.
- Dashboard access should be appropriate to the intended audience.

3.5 Data Requirements

The project requires employee-related information from the available HR datasets.

The main data categories include:

Data Category	Required Information	Analytical Purpose
Employee Information	Employee ID, age, gender, department, job role	Workforce distribution and demographics.
Attrition Information	Attrition status	Measuring employee turnover.
Compensation	Salary, salary hikes	Compensation analysis.
Work Conditions	Overtime, business travel, distance from home, work-life balance	Understanding workplace conditions.

Satisfaction	Job satisfaction, environment satisfaction, relationship satisfaction	Evaluating employee experience.
Performance	Self-rating, manager rating, performance evaluation	Analyzing employee performance.
Training	Training opportunities and completed training sessions	Evaluating employee development.
Career Growth	Promotions and years since last promotion	Understanding career progression.
Employment History	Years at company, years in current role	Analyzing employee tenure and retention.

Data Validation Requirements:

Before analysis, the datasets must be checked for:

- Missing and duplicate records.
- Incorrect or inconsistent data types.
- Unique Employee ID identification.
- Differences in employee counts across tables.
- Multiple records per employee in evaluation tables.
- Inconsistencies in training-related fields.
- Correct relationships between datasets.

3.6 Dashboard Requirements

The final Power BI solution must contain four main dashboard pages.

Page 1: HR Overview & Attrition

- Total Employees.
- Total Attrition.
- Attrition Rate.
- Average Age.
- Workforce distribution by department, job role, gender, and age.
- Attrition analysis by department, job role, and tenure.

Page 2: Satisfaction & Work Environment

- Average Job Satisfaction.
- Average Environment Satisfaction.
- Average Relationship Satisfaction.
- Satisfaction comparison by attrition status.
- Work-life balance analysis.
- Satisfaction analysis by department and tenure.

Page 3: Compensation & Work Conditions

- Average salary by job role.
- Attrition by salary band.
- Attrition by overtime status.
- Business travel analysis.
- Distance from home analysis.

Page 4: Performance, Training & Career Growth

- Performance rating distribution.
- Self-rating versus manager rating.
- Training frequency analysis.
- Salary increase analysis.
- Promotion history.
- Missing performance evaluation records.

3.7 Business Questions & Expected Outputs

The dashboard must provide answers to the following business questions:

Business Question	Expected Output
What is the overall employee attrition rate?	Total attrition and attrition percentage.
Which departments have the highest attrition rates?	Department-level attrition comparison.
Which job roles are most affected by attrition?	Attrition rate by job role.
Are employees with shorter tenure more likely to leave?	Attrition comparison by tenure groups.
Is low job satisfaction associated with higher attrition?	Satisfaction and attrition comparison.
Do lower salary groups experience higher attrition?	Attrition analysis by salary bands.
Is overtime associated with employee attrition?	Attrition comparison by overtime status.
Does performance vary across departments?	Performance distribution by department.
Is training frequency associated with performance?	Training and performance comparison.
Are employees without performance evaluations more likely to leave?	Evaluation availability and attrition comparison.
Which employee groups require greater retention attention?	Identification of groups with high attrition and unfavorable workplace indicators.

3.8 Requirements Summary

The requirements gathering phase establishes the functional, technical, and business expectations for the HR Analytics project.

The resulting Power BI dashboard should provide an interactive environment for exploring workforce data, monitoring employee attrition, investigating potential retention factors, and evaluating employee performance and career development.

By fulfilling these requirements, the project aims to transform available HR data into reliable and meaningful insights that support informed workforce management and employee retention decisions.